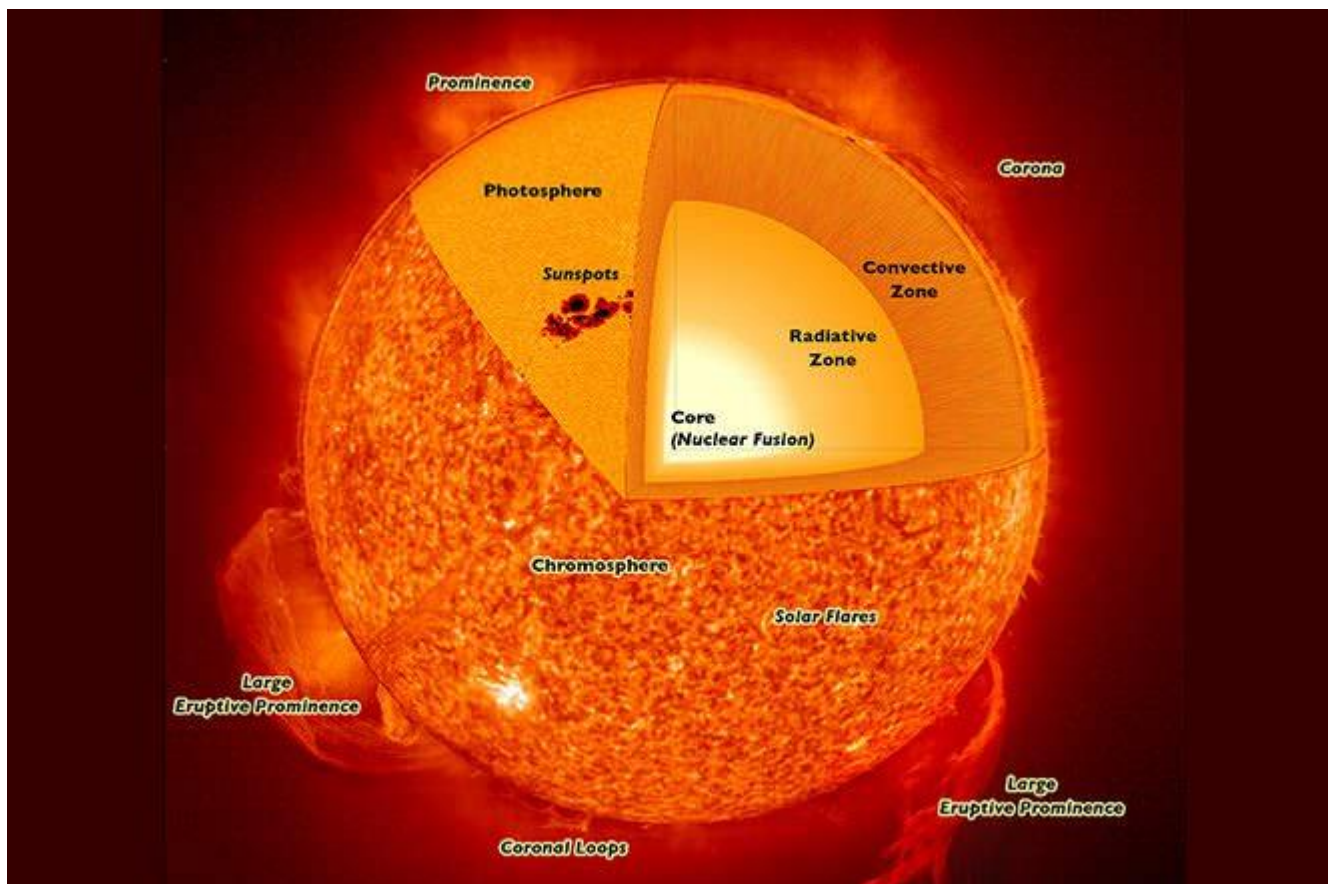


Nuclear Fission and Fusion





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Nuclear fusion and **nuclear fission** are different types of reactions that release energy due to the presence of high-powered atomic bonds between particles found within a nucleus. In fission, an atom is split into two or more smaller, lighter atoms. Fusion, in contrast, occurs when two or more smaller atoms fuse together, creating a larger, heavier atom.

Comparison chart

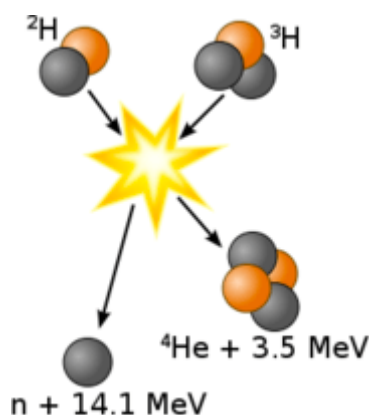
Nuclear Fission versus Nuclear Fusion comparison chart

	Nuclear Fission	Nuclear Fusion
Definition	Fission is the splitting of a large atom into two or more smaller ones.	Fusion is the fusing of two or more lighter atoms into a larger one.
Natural occurrence of the process	Fission reaction does not normally occur in nature.	Fusion occurs in stars, such as the sun.

Nuclear Fission versus Nuclear Fusion comparison chart

	Nuclear Fission	Nuclear Fusion
Byproducts of the reaction	Fission produces many highly radioactive particles.	Few radioactive particles are produced by fusion reaction, but if a fission "trigger" is used, radioactive particles will result from that.
Conditions	Critical mass of the substance and high-speed neutrons are required.	High density, high temperature environment is required.
Energy Requirement	Takes little energy to split two atoms in a fission reaction.	Extremely high energy is required to bring two or more protons close enough that nuclear forces overcome their electrostatic repulsion.
Energy Released	The energy released by fission is a million times greater than that released in chemical reactions, but lower than the energy released by nuclear fusion.	The <u>energy</u> released by fusion is three to four times greater than the energy released by fission.
Nuclear weapon	One class of nuclear weapon is a fission bomb, also known as an atomic bomb or atom bomb.	One class of nuclear weapon is the hydrogen bomb, which uses a fission reaction to "trigger" a fusion reaction.
Energy production	Fission is used in nuclear power plants.	Fusion is an experimental technology for producing power.
Fuel	Uranium is the primary fuel used in power plants.	Hydrogen isotopes (Deuterium and Tritium) are the primary fuel used in experimental fusion power plants.

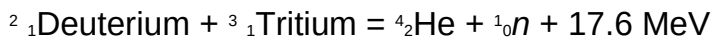
Definitions



Fusion of deuterium with tritium creating helium-4, freeing a neutron, and releasing 17.59 MeV of energy.

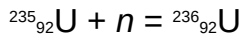
Nuclear fusion is the reaction in which two or more nuclei combine, forming a new element with a higher atomic number (more protons in the nucleus). The

energy released in fusion is related to $E = mc^2$ (Einstein's famous energy-mass equation). On Earth, the most likely fusion reaction is Deuterium–Tritium reaction. Deuterium and Tritium are isotopes of hydrogen.



[Image:Fission-Reaction.svg|thumb|none|Fission Reaction]]

Nuclear fission is the splitting of a massive nucleus into photons in the form of gamma rays, free neutrons, and other subatomic particles. In a typical nuclear reaction involving ${}^{235}_{92}\text{U}$ and a neutron:



followed by



Fission vs. Fusion Physics

Atoms are held together by two of the four fundamental forces of nature: the weak and strong nuclear bonds. The total amount of energy held within the bonds of atoms is called binding energy. The more binding energy held within the bonds, the more stable the atom. Moreover, atoms try to become more stable by increasing their binding energy.

The nucleon of an iron atom is the most stable nucleon found in nature, and it neither fuses nor splits. This is why iron is at the top of the binding energy curve. For atomic nuclei lighter than iron and nickel, energy can be extracted by *combining* iron and nickel nuclei together through nuclear fusion. In contrast, for atomic nuclei heavier than iron or nickel, energy can be released by *splitting* the heavy nuclei through nuclear fission.

The notion of splitting the atom arose from New Zealand-born British physicist Ernest Rutherford's work, which also led to the discovery of the proton.

Conditions for Fission and Fusion

Fission can only occur in large isotopes that contain more neutrons than protons in their nuclei, which leads to a slightly stable environment. Although scientists don't yet fully understand why this instability is so helpful for fission, the general theory is that the large number of protons create a strong repulsive force between them and that too few or too many neutrons create "gaps" that cause weakening of the nuclear bond, leading to decay (radiation). These large nuclei with more "gaps" can be "split" by the impact of thermal neutrons, so called "slow" neutrons.

Conditions must be right for a fission reaction to occur. For fission to be self-sustaining, the substance must reach critical mass, the minimum amount of mass required; falling short of critical mass limits reaction length to mere microseconds. If critical mass is reached too quickly, meaning too many

neutrons are released in nanoseconds, the reaction becomes purely explosive, and no powerful release of energy will occur.

Nuclear reactors are mostly controlled fission systems that use magnetic fields to contain stray neutrons; this creates a roughly 1:1 ratio of neutron release, meaning one neutron emerges from the impact of one neutron. As this number will vary in mathematical proportions, under what is known as [Gaussian distribution](#), the magnetic field must be maintained for the reactor to function, and [control rods](#) must be used to slow down or speed up neutron activity.

Fusion happens when two lighter elements are forced together by enormous energy (pressure and heat) until they fuse into another isotope and release energy. The energy needed to start a fusion reaction is so large that it takes an atomic explosion to produce this reaction. Still, once fusion begins, it can theoretically continue to produce energy as long as it is controlled and the basic fusing isotopes are supplied.

The most common form of fusion, which occurs in stars, is called "D-T fusion," referring to two hydrogen isotopes: [deuterium](#) and [tritium](#). Deuterium has 2 neutrons and tritium has 3, more than the one proton of hydrogen. This makes the fusion process easier as only the charge between two protons needs to be overcome, because fusing the neutrons and the proton requires overcoming the natural repellent force of like-charged particles (protons have a positive charge, compared to neutrons' lack of charge) and a temperature — for an instant — of close to 81 million degrees Fahrenheit for D-T fusion (45 million [Kelvin](#) or slightly less in Celsius). For comparison, the sun's core temperature is roughly 27 million F (15 million C).^[1]

Once this temperature is reached, the resulting fusion has to be contained long enough to generate [plasma](#), one of the four states of matter. The result of such containment is a release of energy from the D-T reaction, producing helium (a noble gas, inert to every reaction) and spare neutrons than can "seed" hydrogen for more fusion reactions. At present, there are no secure ways to induce the initial fusion temperature or contain the fusing reaction to achieve a steady plasma state, but [efforts are ongoing](#).

A third type of reactor is called a breeder reactor. It works by using fission to create plutonium that can seed or serve as fuel for other reactors. Breeder reactors are used extensively in France, but are prohibitively expensive and require significant security measures, as the output of these reactors can be used for making nuclear weapons as well.

Chain Reaction

Fission and fusion nuclear reactions are chain reactions, meaning that one nuclear event causes at least one other nuclear reaction, and typically more. The result is an increasing cycle of reactions that can quickly become uncontrolled. This type of nuclear reaction can be multiple splits of heavy isotopes (e.g. ²³⁵U) or the merging of light isotopes (e.g. ²H and ³H).

Fission chain reactions happen when neutrons bombard unstable isotopes. This type of "impact and scatter" process is difficult to control, but the initial conditions are relatively simple to achieve. A fusion chain reaction develops only under extreme pressure and temperature conditions that remain stable by the energy released in the fusion process. Both the initial conditions and stabilizing fields are very difficult to carry out with current technology.

Energy Ratios

Fusion reactions release 3-4 times more energy than fission reactions. Although there are no Earth-based fusion systems, the sun's output is typical of fusion energy production in that it constantly converts hydrogen isotopes into helium, emitting spectra of light and heat. Fission generates its energy by breaking down one nuclear force (the strong one) and releasing tremendous amounts of heat than are used to heat water (in a reactor) to then generate energy (electricity). Fusion overcomes 2 nuclear forces (strong and weak), and the energy released can be used directly to power a generator; so not only is more energy released, it can also be harnessed for more direct application.

Nuclear Energy Use

The first experimental nuclear reactor for energy production began operating in Chalk River, Ontario, in 1947. The first nuclear energy facility in the U.S., the Experimental Breeder Reactor-1, was launched shortly thereafter, in 1951; it could light 4 bulbs. Three years later, in 1954, the U.S. launched its first nuclear submarine, the U.S.S. Nautilus, while the U.S.S.R. launched the world's first nuclear reactor for large-scale power generation, in Obninsk. The U.S. inaugurated its nuclear power production facility a year later, lighting up Arco, Idaho (pop. 1,000).

The first commercial facility for energy production using nuclear reactors was the Calder Hall Plant, in Windscale (now Sellafield), Great Britain. It was also the site of the first nuclear-related accident in 1957, when a fire broke out due to radiation leaks.

The first large-scale U.S. nuclear plant opened in Shippingport, Pennsylvania, in 1957. Between 1956 and 1973, nearly 40 power production nuclear reactors were launched in the U.S., the largest being Unit One of the Zion Nuclear Power Station in Illinois, with a capacity of 1,155 megawatts. No other reactors ordered since have come online, though others were launched after 1973.

The French launched their first nuclear reactor, the Phénix, capable of producing 250 megawatts of power, in 1973. The most powerful energy-producing reactor in the U.S. (1,315 MW) opened in 1976, at Trojan Power Plant in Oregon. By 1977, the U.S. had 63 nuclear plants in operation, providing 3% of the nation's energy needs. Another 70 were scheduled to come online by 1990.

Unit Two at Three Mile Island suffered a partial meltdown, releasing inert gases ([xenon](#) and [krypton](#)) into the environment. The anti-nuclear movement gained strength from the fears the incident caused. Fears were fueled even more in 1986, when Unit 4 at the Chernobyl plant in Ukraine suffered a runaway nuclear reaction that exploded the facility, spreading radioactive material throughout the area and a large part of Europe. During the 1990s, Germany and especially France expanded their nuclear plants, focusing on smaller and thus more controllable reactors. China launched its first 2 nuclear facilities in 2007, producing a total of 1,866 MW.

Although nuclear energy ranks third behind coal and hydropower in global wattage produced, the push to close nuclear plants, coupled with the increasing costs to build and operate such facilities, has created a pull-back on the use of nuclear energy for power. France leads the world in percentage of electricity produced by nuclear reactors, but in Germany, solar has overtaken nuclear as an energy producer.

The U.S. still has over 60 nuclear facilities in operation, but ballot initiatives and reactor ages have closed plants in Oregon and Washington, while dozens more are targeted by protesters and environmental protection groups. At present, only China appears to be expanding its number of nuclear plants, as it seeks to reduce its heavy dependence on coal (the major factor in its extremely high pollution rate) and seek an alternative to importing oil.

Concerns

The fear of nuclear energy comes from its extremes, as both a weapon and power source. Fission from a reactor creates waste material that is inherently dangerous (see more below) and could be suitable for [dirty bombs](#). Though several countries, such as Germany and France, have excellent track records with their nuclear facilities, other less positive examples, such as those seen in [Three Mile Island](#), [Chernobyl](#), and [Fukushima](#), have made many reluctant to accept nuclear energy, [even though it is much safer than fossil fuel](#). Fusion reactors could one day be the affordable, plentiful energy source that is needed, but only if the extreme conditions needed for creating fusion and managing it can be solved.

Nuclear Waste

The byproduct of fission is radioactive waste that takes thousands of years to lose its dangerous levels of radiation. This means that nuclear fission reactors must also have safeguards for this waste and its transport to uninhabited storage or dump sites. For more information on this, [read about the management of radioactive waste](#).

Natural Occurrence

In [nature](#), fusion occurs in stars, such as the sun. On Earth, nuclear fusion was first achieved in the creation of the [hydrogen bomb](#). Fusion has also

been used in different experimental devices, often with the hope of producing energy in a controlled fashion.

On the other hand, fission is a nuclear process that does not normally occur in nature, as it requires a large mass and an incident neutron. Even so, there have been examples of nuclear fission in natural reactors. This was discovered in 1972 when uranium deposits from an Oklo, Gabon, mine were found to have once sustained a natural fission reaction some 2 billion years ago.

Effects

In brief, if a fission reaction gets out of control, either it explodes or the reactor generating it melts down into a large pile of radioactive slag. Such explosions or meltdowns release tons of radioactive particles into the air and any neighboring surface (land or water), contaminating it every minute the reaction continues. In contrast, a fusion reaction that loses control (becomes unbalanced) slows down and drops temperature until it stops. This is what happens to stars as they burn their hydrogen into helium and lose these elements over thousands of centuries of expulsion. Fusion produces little radioactive waste. If there is any damage, it will happen to the immediate surroundings of the fusion reactor and little else.

It is far safer to use fusion to produce [power](#), but fission is used because it takes less energy to split two atoms than it does to fuse two atoms. Also, the technical challenges involved in controlling fusion reactions have not been overcome yet.

Use of Nuclear Weapons

All nuclear weapons require a nuclear fission reaction to work, but "pure" fission bombs, those that use a fission reaction alone, are known as atomic, or atom, bombs. [Atom bombs](#) were first tested in New Mexico in 1945, during the height of [World War II](#). In the same year, the United States used them as a weapon in Hiroshima and Nagasaki, Japan.

Since the atom bomb, most of the nuclear weapons that have been proposed and/or engineered have enhanced fission reaction(s) in one way or another (e.g., see [boosted fission weapon](#), [radiological bombs](#), and [neutron bombs](#)). Thermonuclear weaponry — a weapon that uses both fission *and* hydrogen-based fusion — is one of the better-known weapon advancements. Though the notion of a [thermonuclear weapon](#) was proposed as early as 1941, it was not until the early 1950s that the hydrogen bomb (H-bomb) was first tested. Unlike atom bombs, hydrogen bombs have *not* been used in warfare, only tested (e.g., see [Tsar Bomba](#)).

To date, no nuclear weapon makes use of nuclear fusion alone, though governmental defense programs have put considerable research into such a possibility.

Cost

Fission is a powerful form of energy production, but it comes with built-in inefficiencies. The nuclear fuel, usually [Uranium-235](#), is expensive to mine and purify. The fission reaction creates [heat](#) that is used to boil water for steam to turn a turbine that generates electricity. This transformation from heat energy to electrical energy is cumbersome and expensive. A third source of inefficiency is that clean-up and storage of nuclear waste is very expensive. Waste is radioactive, requiring proper disposal, and security must be tight to ensure public safety.

For fusion to occur, the atoms must be confined in the [magnetic field](#) and raised to a temperature of 100 [million Kelvin](#) or more. This takes an enormous amount of energy to initiate fusion (atom bombs and lasers are thought to provide that "spark"), but there's also the need to properly contain the plasma field for long-term energy production. Researchers are still trying to overcome these challenges because fusion a safer and more powerful energy production system than fission, meaning it would ultimately cost less than fission.

References

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